

Lecture 7: Profit Maximization under Perfect Competition and Monopoly



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and Society

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Lecture overview

- Alternative Market Structures
- Perfect Competition
- Monopoly
- Contestable Markets

Alternative Market Structures

Firm Behaviour

- What determines how firms behave in a marketplace?
- Why do some products see constant promotion and development while others remain stagnant?
- Why do prices rise in specific industries but fall in others?
 - Firm behavior is dictated by the degree of competition within an industry.
 - The number of rivals and the ease of entry determine a firm's strategy.

Determinants of Market Competition

1. Number of Firms

- Generally, the more firms there are, the higher the level of competition.
- Numerous firms constantly battle to steal market share and customers from one another.

2. Freedom of Entry and Exit

- **Barriers to Entry:** High start-up costs, patents, or legal regulations can prevent new rivals from joining the market.
- **Impact on Competition:**
 - **Low Barriers:** New firms enter easily, increasing competition.
 - **High Barriers:** Existing firms are protected, restricting the number of competitors.

Determinants of Market Competition

3. Nature of the Product

- **Homogeneous Products:** If products are identical (e.g., raw sugar or gold), firms have little room to gain an advantage other than through price.
- **Product Differentiation:** Unique branding, models, or varieties allow firms to stand out. Enables firms to charge higher prices or secure a larger market share despite rivals.

4. Shape of the Demand Curve (Price Control): The degree of competition is reflected in the firm's Price Elasticity of Demand:

- **Perfectly Elastic (Horizontal):** The firm has zero control; raising prices by even 1% results in losing all sales.
- **Relatively Elastic:** A price increase leads to a large drop in sales (high competition).
- **Relatively Inelastic:** A price increase leads to only a small drop in sales (low competition/high brand loyalty).

Alternative Market Structures

Industries are categorized based on their level of competition, moving from most competitive to least competitive.

- **Perfect Competition (Most Competitive)**
 - Large number of firms.
 - Firms are price takers with no individual market power.
- **Monopolistic Competition (Imperfect)**
 - Quite a lot of firms with freedom of entry.
 - Common in local services (e.g., businesses found in search engines).
- **Oligopoly (Imperfect)**
 - Only a few dominant firms.
 - High barriers to entry; firms have high market share and influence over price and design.
- **Monopoly (Least Competitive)**
 - Only one firm in the industry.
 - No internal competition due to very high barriers to entry.

Features of the four market structures

Type of market	Number of firms	Freedom of entry	Nature of product	Examples	Implications for demand curve faced by firm
Perfect competition					
Monopolistic competition					
Oligopoly					
Monopoly					

Features of the four market structures

Type of market	Number of firms	Freedom of entry	Nature of product	Examples	Implications for demand curve faced by firm
Perfect competition	Very many	Unrestricted	Homogeneous (undifferentiated)	Cabbages, carrots (approximately)	Horizontal: firm is a price taker
Monopolistic competition	Many / several	Unrestricted	Differentiated	Builders, restaurants	Downward sloping, but relatively elastic
Oligopoly	Few	Restricted	1. Undifferentiated 2. Differentiated	Petrol Cars, electrical appliances	Downward sloping. Relatively inelastic (shape depends on reactions of rivals)
Monopoly	One	Restricted or completely blocked	Unique	Local water company, train operators (over particular routes)	Downward sloping: more inelastic than oligopoly. Firm has considerable control over price

Q Under which of the following market structures does Tesco operate?

- A. Perfect competition
- B. Monopolistic competition
- ☒ C. Oligopoly
- D. Monopoly

Perfect Competition

Perfect Competition – The Theoretical Ideal



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- Firms have no power to affect the price.
- The market price is determined solely by the intersection of total market demand and supply.

Perfect Competition – The Theoretical Ideal



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The Four Core Assumptions:

- **Price Takers:**

- Firms are so small relative to the industry.
- A horizontal (perfectly elastic) demand curve at the market price.

- **Freedom of Entry and Exit:**

- No barriers to new firms entering the market. This typically applies in the long run, as setting up takes time.

- **Homogeneous Products:**

- All goods are identical and perfect substitutes.
- No branding or advertising is necessary (or effective).

- **Perfect Knowledge:**

- Producers know all costs, prices, and technologies.
- Consumers are fully aware of price, quality, and availability.

Short-Run Equilibrium in Perfect Competition

The number of firms is fixed because there is no time for new competitors to enter the market.

Price Determination

- The Industry: Price is set by the intersection of Market Demand and Market Supply.
- The Firm: Acts as a price taker, facing a horizontal demand curve at the market price (P_e).
 - Selling above P_e results in zero sales (consumers switch to identical, cheaper rivals).
 - Selling below P_e is unnecessary, as the firm can sell all it produces at the market price.

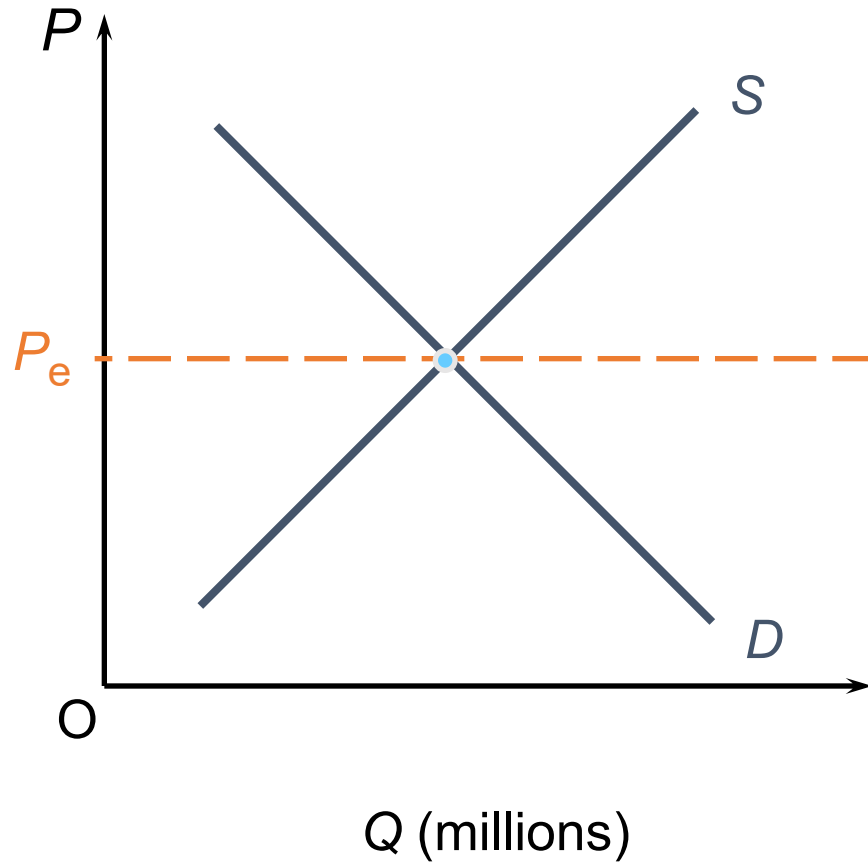
Short-Run Equilibrium in Perfect Competition

Output and Profit Maximization

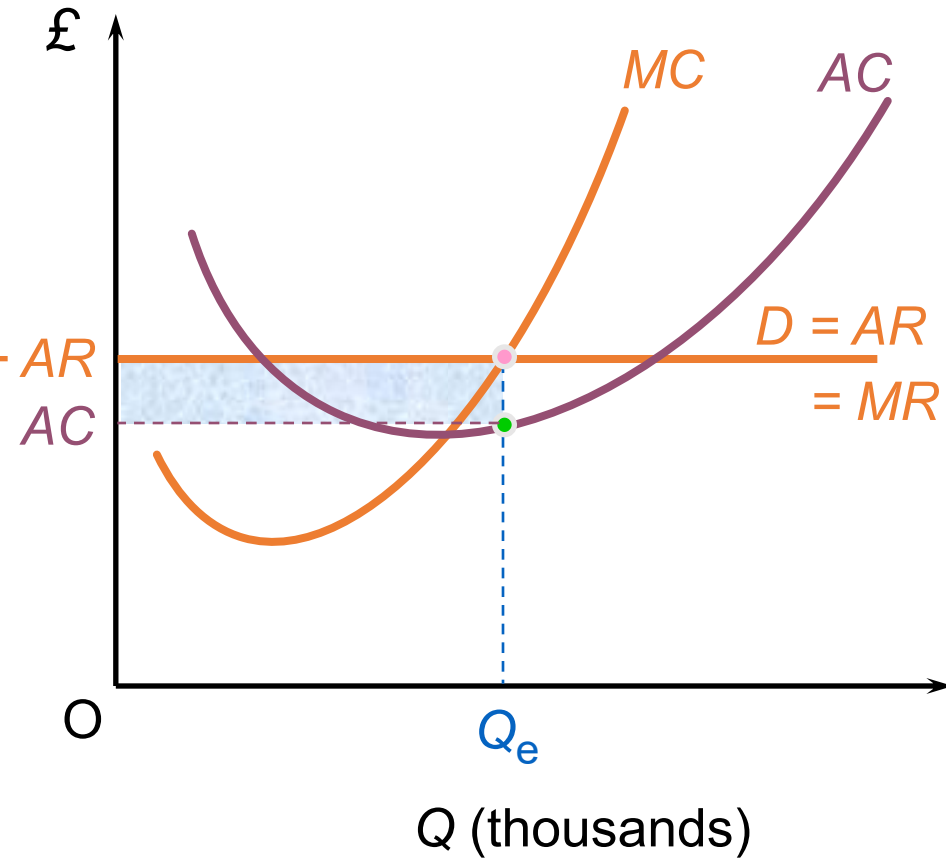
- The firm produces where Marginal Revenue (MR) = Marginal Cost (MC).
- Because the price is constant for the firm, Price = AR = MR.
- Supernormal Profit: Occurs if the Average Revenue (AR) curve is above the Average Cost (AC) curve at the equilibrium output (Q_e).
 - $(AR - AC) \times \text{Quantity}$.
 - Represented as a shaded rectangle on the diagram.

Short-run equilibrium of industry and firm under perfect competition

Firm is a price taker. Price is given by the market.



(a) Industry



(b) Firm

Long-Run Equilibrium in Perfect Competition

In the long run, all factors of production are variable, and firms can freely enter or exit the industry.

The Mechanism of Change

1. Existing supernormal profits attract new firms into the industry.
2. The entry of new firms shifts the Industry Supply Curve to the right.
3. As supply increases, the market price falls.
4. The price continues to fall until supernormal profits are wiped out.

Long-Run Equilibrium in Perfect Competition

The Equilibrium State

- Normal Profit Only: Equilibrium is reached when the price (Demand/AR) just touches the minimum point of the Long-Run Average Cost (LRAC) curve.
- At equilibrium, the following equality holds:

$$\text{LRAC} = \text{AC} = \text{MC} = \text{MR} = \text{AR}$$

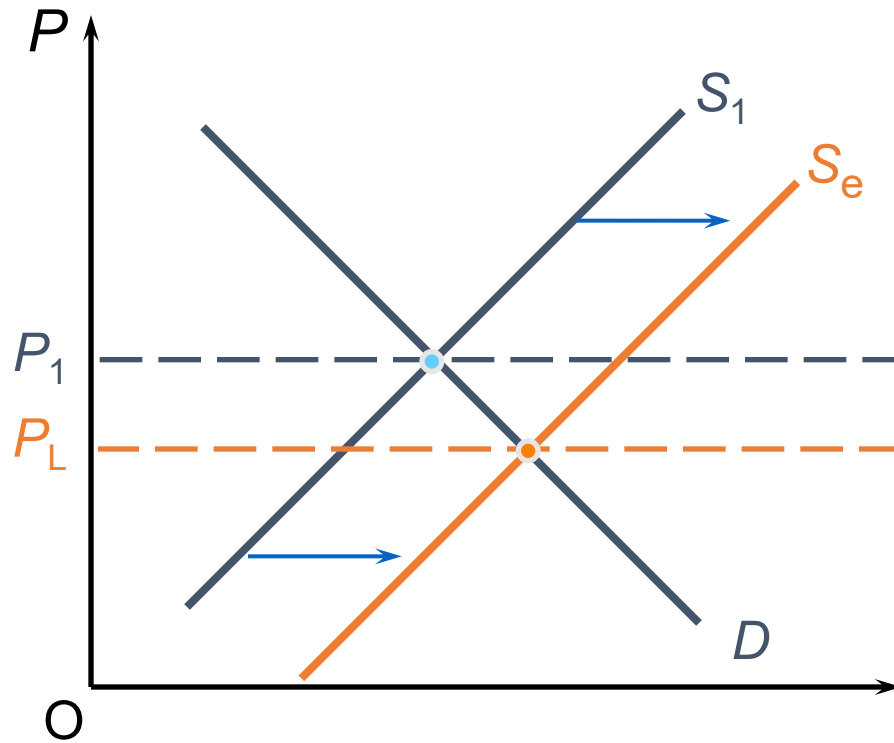
- Firms operate at their most efficient scale (the bottom of the AC curve), and no further incentive exists for firms to enter or leave.

Long-run equilibrium under perfect competition

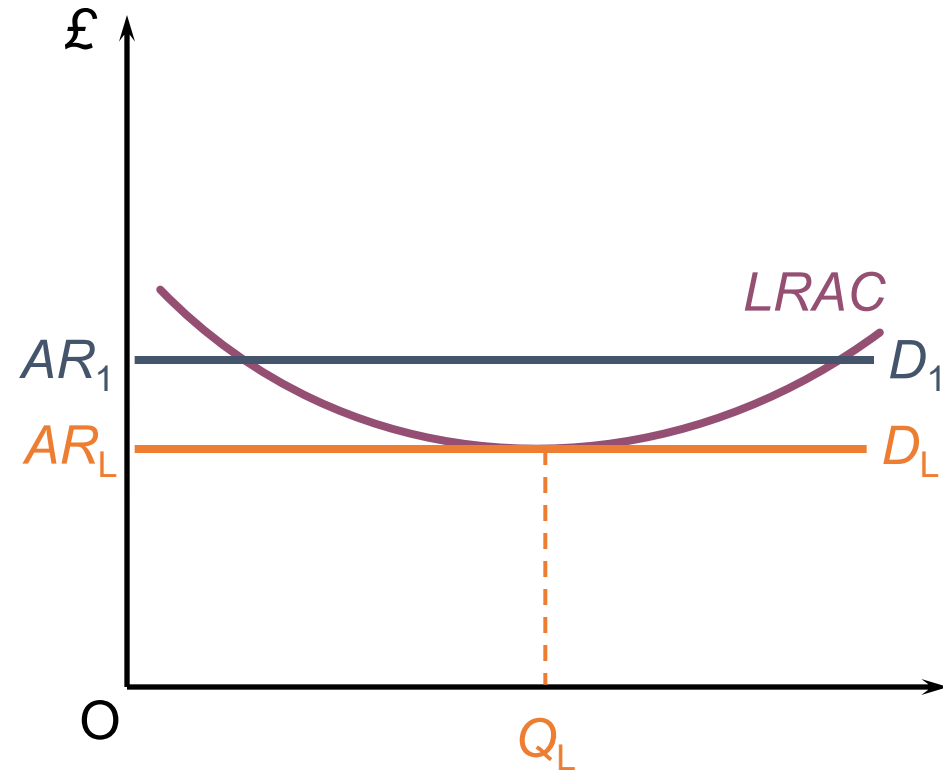
New firms enter.

Profits return
to normal.

ts



(a) Industry



(b) Firm

Monopoly

Monopoly

A monopoly technically exists when there is only **one firm** in an industry, but the definition is often fluid.

- **Industry Boundaries:** Whether a firm is a monopolist depends on how narrowly you define the market (e.g., a monopoly on silk fabric vs. the textile industry).
- **Monopoly Power:** True power depends on the closeness of substitutes from other industries.
 - *Example:* A train company may have a monopoly on rail travel between two cities but still competes with cars, coaches, and planes.

Barriers to Entry (Structural)

To maintain a monopoly, a firm must prevent new rivals from entering the market.

- **Economies of Scale (Natural Monopoly):**
 - Occurs when average costs fall continuously as output increases.
 - Common in industries with high infrastructure costs and low marginal costs (e.g., gas pipelines).
 - *Example:* It is inefficient for two firms to build two separate national grids; one firm can serve everyone at a lower average cost.
- **Economies of Scope:**
 - Cost advantages gained by producing a range of products.
 - *Example:* A pharmaceutical giant shares research, storage, and transport across drugs and toiletries, undercutting single-product specialists.
- **Product Differentiation & Brand Loyalty:**
 - Strong consumer identification with a brand makes it difficult for new entrants to gain a foothold.

Barriers to Entry (Strategic & Legal)

- **Lower Costs for Established Firms:**
 - Incumbents have optimized production, exclusive supplier knowledge, and access to cheaper finance.
- **Control over Inputs or Outlets:**
 - Exclusive Contracts: Signing long-term deals with the best suppliers.
 - Vertical Integration: Buying out the only supplier of a key raw material to block rivals.
- **Legal Protection:**
 - Intellectual Property: Patents and copyrights protect unique processes.
 - Government Action: Licensing (only one firm allowed to operate) or tariffs to block foreign competition.
- **Aggressive Takeovers:**
 - The monopolist may simply buy out any new entrant that shows potential.
 - The mere threat of a takeover can discourage investors from funding new competitors.

Q Which one of the following would not be a barrier to firms entering an industry?

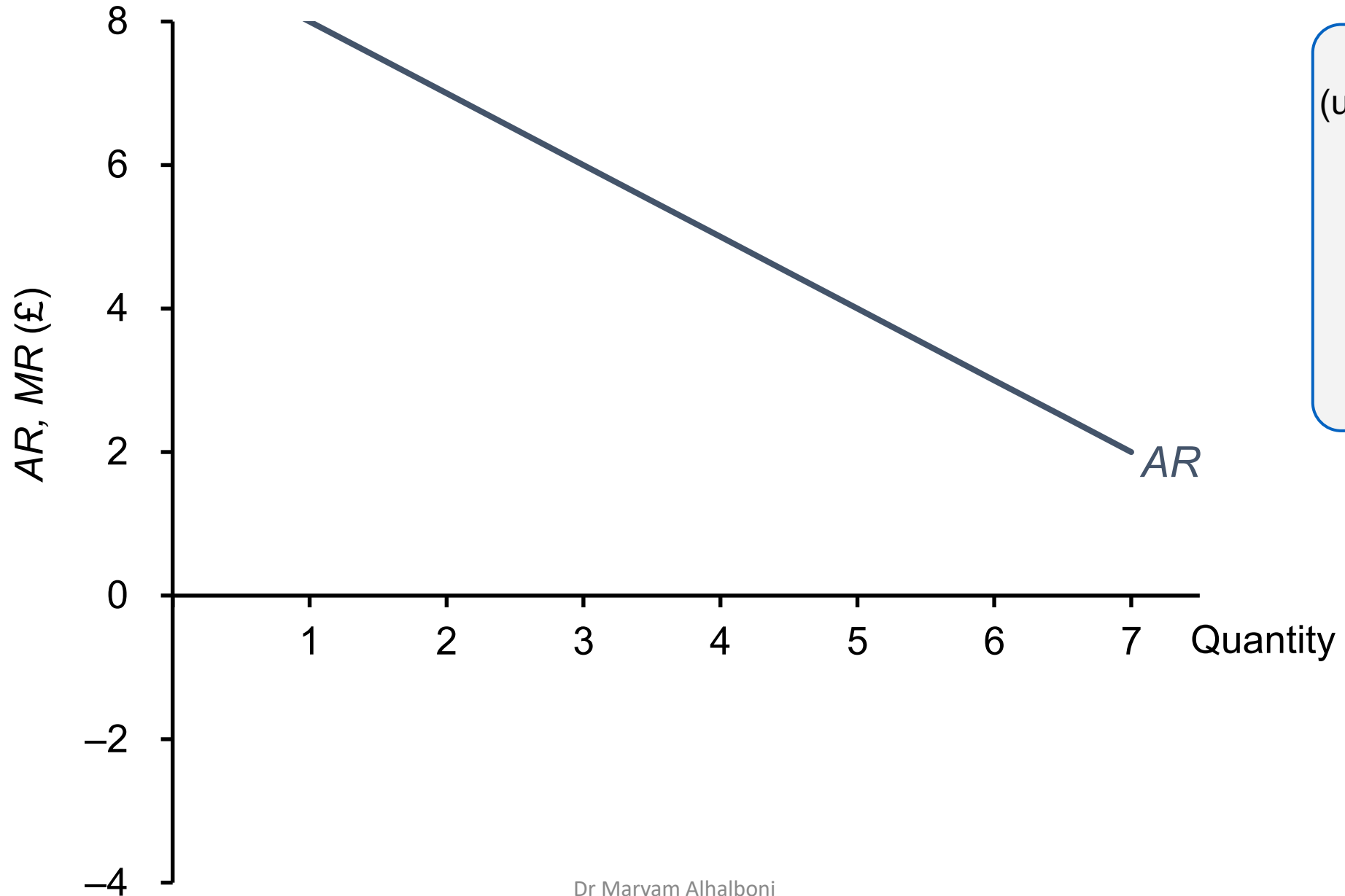
- A.** An upward-sloping long-run average cost curve
- B. Patents on key processes
- C. Substantial economies of scale
- D. Large initial capital costs
- E. The threat of takeover by the existing firm(s)

Monopoly Equilibrium – Price and Output

As the sole provider in the industry, the monopolist faces the entire market's demand.

- **The Industry is the Firm:** The firm's demand curve is identical to the industry demand curve.
- **Price Maker Status:**
 - Unlike perfect competition, the monopolist can choose its price.
 - Because there are no internal rivals, demand is relatively inelastic; consumers must pay the higher price or forgo the product.
- While the monopolist is a price maker, they are still limited by the demand curve, raising the price will inevitably reduce the quantity sold.

AR and *MR* curves for a monopoly



Contestable Markets

The Theory of Contestable Markets

What if the *threat* of competition is just as powerful as competition itself?

- It is not the *actual* number of firms that determines price, but the threat of potential entry.
- **Case Study: University Catering:**
 - A cafe might have a monopoly on campus, but if it provides poor service or high prices, the university can switch to a rival provider.
 - This threat keeps prices reasonable and service high, even if only one firm is currently operating.

Perfectly Contestable Markets

A market is perfectly contestable when entry and exit costs are zero.

- **Behavior:** Firms will keep prices low (earning only normal profit) and maintain maximum efficiency to prevent rivals from entering.
- **Efficiency:** Firms must utilize new technology and economies of scale; otherwise, a more efficient rival will enter and replace them.

Contestability and Natural Monopolies



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- Why stay a monopoly? Some markets only have room for one firm to operate at the Minimum Efficient Scale (economies of scale).
- If a market isn't big enough for two, but entry/exit costs are zero, a new firm will enter if it thinks it is more efficient than the incumbent.
- The existing monopolist must act as if it were in a competitive market to survive.

Questions